

LLM Behaviours in Idealised AI Race Experiments

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ABSTRACT

Behavioural evidence suggests that unsafe development in technological races emerges less from individual risk preferences than from the strategic dynamics of competition: opponent behaviour, relative standing, and early momentum. Whether large language models (LLMs), increasingly deployed as autonomous or semi-autonomous agents, reproduce these observable dynamics in an analogous competitive setting remains open. We extend a prior human behavioural race paradigm, in which paired participants repeatedly chose between Safe and Unsafe development under an uncertain time horizon and treatment-specific maximum risk (10%, 60%, 90%), to LLM agents. The protocol defines two-player and N -player conditions and persona-related prompt framings, while the current exploratory evidence is bounded to the available model-prompt-decoding runs. We compare observed LLM trajectories with published human estimands on three dimensions identified as behaviourally consequential in humans: conditioning on the opponent’s prior move, sensitivity to being ahead versus behind, and path dependence from first-round choice. We also test whether LLMs reproduce the human null result for risk-level treatment or instead show explicit sensitivity to stated probabilities. The abstract is provisional: the final version will report the frozen audit and behavioural results and will not treat LLM outputs as a replacement for participant-level human data or as evidence of subjective risk preferences.

KEYWORDS

large language models, multi-agent systems, AI safety, repeated games, behavioural game theory, competitive risk-taking

1 INTRODUCTION

Rapid progress in artificial intelligence, under the assumption that the first superintelligent AI will be deeply transformative, has intensified competition among companies, research laboratories, and governments seeking technological and strategic advantage. This competition is often characterised as an AI race, in which participants benefit from reaching increasingly capable systems before their competitors. Earlier theoretical work suggests that such winner-takes-most dynamics may encourage developers to reduce investment in safety when safety measures slow development [4]. This competitive advantage produces a collective-action problem: although all participants benefit from responsible development, each has an individual incentive to move faster by investing less in safety [5]. The idealised AI-race model provides a controlled way to study this tension between technological progress and safety. For example, Han et al. [10] modelled AI development as a repeated race in which agents choose between SAFE and UNSAFE strategies. Unsafe development provides a speed advantage but increases the

probability of a harmful outcome. Therefore, the AI race must be understood not merely as a series of isolated decisions, but as complex strategic interactions among competing actors.

Formal game-theoretic models are useful because systematic data on real AI-development decisions remain limited. Recent behavioural evidence indicates that choices may emerge not simply from fixed attitudes towards risk but from the evolving strategic position of participants. In an idealised AI-race experiment with human participants, Domingos and Han [8] found that falling behind a competitor and observing the competitor choose unsafe development increased the likelihood of subsequent unsafe choices. These findings highlight the importance of conditional behaviour and strategic imitation when studying safety decisions in competitive environments.

Because experiments with human participants are costly and difficult to scale to long-horizon interactive settings, LLMs provide a practical candidate for studying these interactions at scale [1, 3]. We first use LLMs to reproduce the two-player AI-race experiment developed by Domingos and Han [8]. The models receive the same rules, payoffs, risk levels, and information as the human participants. At the end of the race, the player with the greatest progress wins the final prize, while tied winners share it. Only winners or tied winners face the accumulated private setback risk, which may cause them to lose all their benefits [8, 10]. This setting allows comparison with published human estimands under similar game conditions, following earlier work that evaluates LLMs through human experiments and repeated games [1, 2]. We test whether AI models show the main patterns found in the human experiment, including responses to private risk, the opponent’s previous action, relative progress, and first-round choices [8]. We also check whether the models understand the rules and correctly update progress, risk, and payoffs before treating their actions as strategic behaviour. This validation is important because similarity in average responses does not necessarily show that LLMs follow the same decision process as humans [6].

The two-player game provides a clear starting point, but many competitive settings involve more than two actors [4, 5]. In an N -player race, each actor must respond to several competitors rather than a single opponent. The number and structure of these interactions may change incentives for risk-taking and safety [7, 9]. More players also create new sources of pressure. A model may react to its current rank, its distance from the leader, the number of competitors ahead, or the overall level of Unsafe behaviour in the group. Previous research shows that trailing competitors often accept more risk to catch up, whereas leaders tend to act more cautiously [11, 12]. Unsafe choices may also spread when models copy one another or try to avoid falling behind. For these reasons, patterns found in a two-player game may not remain the same as the group becomes larger.

We extend the experiment to an interactive N -player engine while keeping the main structure of the idealised AI-race model [10]. In each round, all models choose Safe or Unsafe at the same time. Safe and Unsafe choices retain their differences in cost, progress, immediate benefit, and accumulated private risk. The models' choices jointly affect their payoffs, progress, ranking, and chances of winning the final prize. Before the next decision, each agent receives updated information about the race, allowing its behaviour to change as the group develops. The current exploratory evidence covers $N = 3$; matched $N = 4$ and $N = 5$ runs remain planned, so no group-size effect is estimated here. This extension is exploratory because there is not yet a matching N -player human dataset.

Figure 1 summarises this mechanism across both group sizes, and Figure 2 summarises the three persona conditions used to vary how each company's executive is framed.

This study aims to determine whether large language model (LLM) agents can reproduce observable patterns of safety behaviour in an idealised AI race and whether these patterns remain stable as the competitive setting changes. We use a staged approach because an output does not by itself show that an agent has followed the rules or correctly represented the evolving state of the race. The analysis therefore begins with task-validity checks, proceeds to the two-player human benchmark, and then considers the exploratory N -player extension. This paper addresses the following research questions:

- Can LLM agents reliably execute the rules and state transitions of the AI-race game?
- How are private risk, prior actions, race position, and early choices associated with UNSAFE play?
- How stable are these behavioural patterns across models, prompts, personas, and decoding protocols?
- How do group size and multiplayer race state relate to UNSAFE play in exploratory N -player settings?

2 PRELIMINARIES

We first specify the two-player game reproduced from Domingos and Han [8] (§2.1), then its generalisation to $N > 2$ companies (§2.2). Every quantity that is not restated in §2.2, namely the horizon, prize split, and private-risk formula, carries over unchanged from the two-player game; only the stage payoff changes shape (Figure 1).

2.1 Two-player game

Two companies $i \in \{1, 2\}$ play a repeated race. In every round t , both companies simultaneously choose an action $a_i^t \in \{S, U\}$ (Safe, Unsafe) and commit to it before either learns the other's choice for that round; only once both actions are committed does the round resolve (Figure 1a). Resolution has two parts. First, Safe advances the company's cumulative progress by $\sigma(S) = 1.0$ and Unsafe advances it by $\sigma(U) = 1.5$, so Unsafe always closes the race faster; progress accumulates as $P_i^t = P_i^{t-1} + \sigma(a_i^t)$. Second, the round pays a stage payoff $\pi_i^t = \pi(a_i^t, a_{-i}^t)$, read off the fixed matrix (own action by row, opponent action by column, Figure 1b):

$$\pi = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}. \quad (1)$$

Unsafe strictly dominates Safe at the stage-game level (row 2 exceeds row 1 in both columns): the only cost of Unsafe development enters later, through accumulated private risk rather than through the round payoff itself.

The race lasts a minimum of 5 rounds; after every completed round from round 5 onward it terminates with probability 0.2, giving an expected horizon $\mathbb{E}[T] = 9$. Neither company is ever told the horizon in advance, so a company can never condition its action on knowing the current round is the last one (Figure 1a). Termination then resolves in two further steps. First, the company with the higher cumulative progress P_i^T is declared the winner of a prize $B = 100$; if progress is exactly tied, the two companies split it 50/50. Second, and only for that winner or tied winners, an accumulated private setback risk is drawn:

$$q_i(T) = p_r^{\max} \cdot \frac{n_i^U(T)}{T}, q_i(T) \in [0, 1], p_r^{\max} \in \{0.10, 0.60, 0.90\}, \quad (2)$$

where $n_i^U(T)$ counts company i 's Unsafe choices over the whole race and $p_r^{\max}$ is a treatment-level constant, assigned once per race, that we vary across three risk levels. If the draw against $q_i(T)$ fails, that company's entire race payoff, every stage payoff it accumulated plus the prize, is zeroed; if it succeeds, the company keeps both. A company that ends behind is never subject to this draw at all: it keeps its accumulated stage payoffs regardless of how much Unsafe it played, but receives no share of the prize (Figure 1a). This asymmetry, in which risk is built by both companies but realised only against whoever is ahead, is what makes Unsafe development a genuine gamble rather than a free speed advantage.

2.2 N -player game

The N -player game keeps every element of §2.1 unchanged (simultaneous sealed actions, progress increments $\sigma(S) = 1.0/\sigma(U) = 1.5$, the minimum-5-round hidden horizon, terminal prize $B = 100$, and the risk formula, Equation 2), but is played by $N \geq 2$ companies indexed by $i \in \{1, \dots, N\}$ instead of exactly two, and the prize at round T splits evenly among however many companies share the highest progress rather than only two. The engine supports $N \in \{2, 3, 4, 5\}$; the current output covers $N = 3$, while matched $N = 4$ and $N = 5$ conditions remain planned (Figure 1c).

With more than two companies there is no single opponent to index a matrix against, so Equation 1 generalises to the N -player stage-payoff rule of Han et al. [10] (their Appendix B), evaluated at $k^t \in \{0, 1, \dots, N\}$, the number of companies (out of N) choosing Safe in round t . Therefore, every Safe-choosing company earns

$$\pi_{\text{Safe}}(k^t) = \frac{b}{k^t + s(N - k^t)} - c, \quad (3)$$

and every Unsafe-choosing company earns

$$\pi_{\text{Unsafe}}(k^t) = s \cdot \frac{b}{k^t + s(N - k^t)}, \quad (4)$$

with cost $c = 1$, per-round benefit $b = 4$, and speed $s = 1.5$ fixed across all group sizes (the per-round benefit b is distinct from terminal prize B , which is paid once, at the end of the race, not every round). Both payoffs depend on the group's joint action counts, k^t and $N - k^t$, not on any single rival's action, which is the structural difference from the two-player game: a company's stage payoff

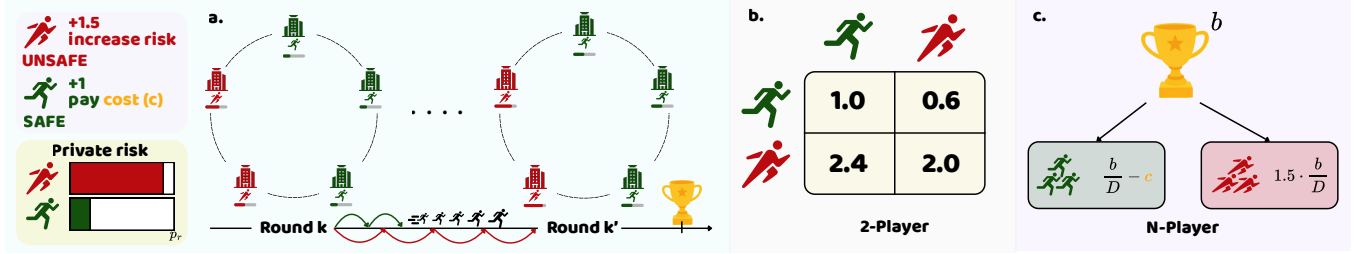


Figure 1: Overview of the AI-race mechanism shared by the two-player and N -player games. (a) Each round, every company advances along the track by choosing Safe (green, +1 progress, pays cost c) or Unsafe (red, +1.5 progress, increases private setback risk); the race ends after a hidden number of rounds, at which point terminal prize B is awarded to the progress leader or divided among tied leaders. (b) The two-player stage-payoff matrix: own Safe/opponent Safe pays 1.0, own Safe/opponent Unsafe pays 0.6, own Unsafe/opponent Safe pays 2.4, and own Unsafe/opponent Unsafe pays 2.0. (c) The N -player stage-payoff rule: if k of N companies choose Safe, $D = k + s(N - k)$; each Safe-choosing company earns $b/D - c$ and each Unsafe-choosing company earns $1.5 \cdot b/D$, where b is the per-round benefit and is distinct from terminal prize B .

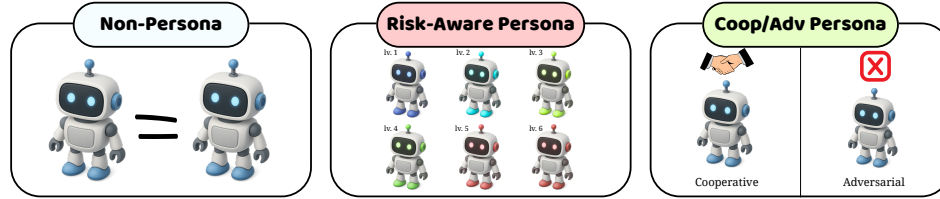


Figure 2: Overview of the three persona-related framing families used to describe each company’s executive. The canonical baseline contains no persona sentence; a separate neutral placebo uses a length-matched neutral sentence. The risk-aware family contains six levels adapted from the Eckel–Grossman gamble scale. The cooperative/adversarial family frames each seat as believing rival firms do better through mutual restraint or only at each other’s expense. These are prompt conditions, not measured psychological traits.

now moves with how many of its rivals, not which one, chose each action.

2.3 Playground Simulation

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